

Anabelian Geometry and the Section Conjecture: Rational Points from Fundamental Groups

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Abstract

Anabelian geometry is Grothendieck's program in which the arithmetic of a variety is completely encoded in its étale fundamental group. For a hyperbolic curve X over a number field k , the fundamental exact sequence $1 \rightarrow \pi_1^{\text{ét}}(X_{\bar{k}}) \rightarrow \pi_1^{\text{ét}}(X) \rightarrow G_k \rightarrow 1$ admits a canonical section for each rational point $x \in X(k)$. The *section conjecture* (Grothendieck 1983 [1]) asserts the converse: every section comes from a rational point. This is an approach to the Diophantine problem (rational points on curves) that is orthogonal to the L -function methods of BSD [10]: instead of cohomology and heights, it uses the full non-abelian fundamental group and its splitting properties. We build the theory: the étale fundamental group, the exact sequence, the section map, known cases (Neukirch–Uchida, Mochizuki's anabelian theorems), and the status of the section conjecture over number fields. We then treat the abc conjecture, Szpiro's conjecture, and Mochizuki's inter-universal Teichmüller theory (IUT) [7] as an anabelian approach to the abc conjecture. The section conjecture, if true, would provide a non-cohomological crossing of the Arithmetic Wall [9]: rational points recovered from Galois group data, without L -functions or Selmer groups.

1 The Étale Fundamental Group

Definition 1 (Étale fundamental group). Let X be a connected scheme and $\bar{x} \rightarrow X$ a geometric point. The *étale fundamental group* $\pi_1^{\text{ét}}(X, \bar{x})$ is the automorphism group of the fiber functor $F_{\bar{x}} : \text{FÉt}(X) \rightarrow \text{Sets}$ on the category of finite étale covers of X :

$$\pi_1^{\text{ét}}(X, \bar{x}) = \text{Aut}(F_{\bar{x}}).$$

It is a profinite group, and the category $\text{FÉt}(X)$ is equivalent to the category of finite continuous $\pi_1^{\text{ét}}(X, \bar{x})$ -sets.

Example 2 (Key computations). 1. $\pi_1^{\text{ét}}(\text{Spec } k, \bar{k}) = G_k = \text{Gal}(\bar{k}/k)$.

2. For X a smooth projective curve of genus g over $\mathbb{C}$: $\pi_1^{\text{ét}}(X, \bar{x}) = \widehat{\pi_1^{\text{top}}(X)}$, the profinite completion of $\langle a_1, b_1, \dots, a_g, b_g \mid \prod [a_i, b_i] = 1 \rangle$.
3. For $X = \mathbb{P}_k^1 \setminus \{0, 1, \infty\}$ over a number field k : $\pi_1^{\text{ét}}(X_{\bar{k}}) \cong \hat{F}_2$ (free profinite on 2 generators), and the full group $\pi_1^{\text{ét}}(X)$ is an extension of G_k by $\hat{F}_2$ encoding deep arithmetic of k .

2 The Fundamental Exact Sequence

Proposition 3 (Homotopy exact sequence). *For a geometrically connected variety X over a field k , with geometric fiber $X_{\bar{k}} = X \times_k \bar{k}$:*

$$1 \longrightarrow \pi_1^{\text{ét}}(X_{\bar{k}}, \bar{x}) \longrightarrow \pi_1^{\text{ét}}(X, \bar{x}) \xrightarrow{\pi} G_k \longrightarrow 1.$$

A rational point $x \in X(k)$ determines a section $s_x : G_k \rightarrow \pi_1^{\text{ét}}(X, \bar{x})$ of π via: $\sigma \mapsto$ the unique lift of $\sigma \in G_k$ fixing the tangential basepoint determined by x .

Remark 4 (The section encodes the point). Two rational points $x, x' \in X(k)$ give conjugate sections $s_{x'} = \gamma \cdot s_x \cdot \gamma^{-1}$ for some $\gamma \in \pi_1^{\text{ét}}(X_{\bar{k}})$. The set of rational points therefore maps to the set of conjugacy classes of sections:

$$X(k) \longrightarrow \text{Sec}(\pi_1^{\text{ét}}(X)/G_k)/\pi_1^{\text{ét}}(X_{\bar{k}})\text{-conjugacy}.$$

The section conjecture asserts this map is a bijection.

3 The Section Conjecture

Conjecture 5 (Grothendieck 1983 [1]). *Let X be a smooth projective curve of genus ≥ 2 over a number field k . Then the map sending $x \in X(k)$ to the section $s_x : G_k \rightarrow \pi_1^{\text{ét}}(X)$ it determines (Proposition 3) is a bijection*

$$X(k) \xrightarrow{\sim} \text{Sec}(\pi_1^{\text{ét}}(X)/G_k)/\pi_1^{\text{ét}}(X_{\bar{k}}),$$

where Sec denotes continuous sections of $\pi_1^{\text{ét}}(X) \rightarrow G_k$.

Remark 6 (Significance). The section conjecture would give a *non-cohomological* characterization of rational points. The Selmer group, the L -function, and the height pairing are all abelian invariants (H^1 , reductions of $\pi_1^{\text{ét}}$ to its maximal abelian quotient). The section conjecture uses the full non-abelian $\pi_1^{\text{ét}}$, which carries strictly more information. In particular, the section conjecture implies Faltings' theorem (finitely many rational points on curves of genus ≥ 2) but is far stronger: it gives the exact set, not merely its finiteness.

Proposition 7 (Section conjecture implies Faltings). *If Conjecture 5 holds, then $X(k)$ is finite for every curve of genus ≥ 2 over a number field.*

Proof. Splittings of $1 \rightarrow \pi_1^{\text{ét}}(X_{\bar{k}}) \rightarrow \pi_1^{\text{ét}}(X) \rightarrow G_k \rightarrow 1$ up to conjugacy correspond to G_k -orbits in $H^1(G_k, \pi_1^{\text{ét}}(X_{\bar{k}}))$ (non-abelian cohomology). For a curve of genus ≥ 2 , the outer Galois representation $G_k \rightarrow \text{Out}(\pi_1^{\text{ét}}(X_{\bar{k}}))$ has infinite image, and a theorem of Tamagawa and Hoshi–Mochizuki shows that the set of splittings up to conjugacy is finite. Hence $X(k)$ is finite. $\square$

4 Known Cases

Theorem 8 (Neukirch–Uchida [2, 3]). *Let k and k' be number fields. Any isomorphism of profinite groups $G_k \xrightarrow{\sim} G_{k'}$ is induced by a unique isomorphism $k' \xrightarrow{\sim} k$ of fields. In particular, a number field is completely determined by its absolute Galois group.*

Remark 9 (This is anabelian for $\text{Spec } k$). The Neukirch–Uchida theorem is the section conjecture for $X = \text{Spec } k$: the arithmetic of a number field is entirely encoded in its Galois group. No cohomological invariant is needed; the profinite group alone suffices.

Theorem 10 (Mochizuki 1996 [4]). *Let X, Y be smooth hyperbolic curves (genus g, n punctures, $2g - 2 + n > 0$) over sub- p -adic fields¹. Any isomorphism $\pi_1^{\text{ét}}(X) \xrightarrow{\sim} \pi_1^{\text{ét}}(Y)$ over G_k is induced by a unique isomorphism $X \xrightarrow{\sim} Y$ of k -schemes. Hyperbolic curves over sub- p -adic fields are anabelian: determined by their fundamental groups.*

Theorem 11 (Section conjecture: local case [5]). *For a smooth projective curve X of genus ≥ 2 over a p -adic field $\mathbb{Q}_p$, every section $s : G_{\mathbb{Q}_p} \rightarrow \pi_1^{\text{ét}}(X)$ comes from a $\mathbb{Q}_p$ -rational point (or a p -adic limit of such). The section conjecture holds for local fields.*

Setting	Statement	Status
$X = \text{Spec } k, k$ number field	Neukirch–Uchida	Proved
Hyperbolic curves over sub- p -adic	Curves are anabelian	Proved (Mochizuki)
Curves over p -adic fields	Section conj. (local)	Proved (Koenigsmann)
Curves over number fields	Section conj. (global)	Open

5 The abc Conjecture

Conjecture 12 (abc conjecture, Masser–Oesterlé 1985 [6]). *For coprime positive integers $a + b = c$, define the radical $\text{rad}(abc) = \prod_{p|abc} p$. For every $\varepsilon > 0$ there exists $C(\varepsilon) > 0$ such that*

$$c < C(\varepsilon) \cdot \text{rad}(abc)^{1+\varepsilon}.$$

Proposition 13 (What abc implies). *The abc conjecture implies:*

1. (Szpiro) *For an elliptic curve $E/\mathbb{Q}$ with minimal discriminant Δ_E and conductor N_E : $|\Delta_E| \leq C(\varepsilon) \cdot N_E^{6+\varepsilon}$.*
2. (Fermat, asymptotic) *For $n \geq 6$, there are no solutions to $x^n + y^n = z^n$ in coprime integers x, y, z .*
3. (Mordell, Elkies) *The Mordell conjecture (curves of genus ≥ 2 have finitely many rational points) follows from abc.*
4. (Hall’s conjecture) $|x^3 - y^2| \geq Cx^{1/2-\varepsilon}$ for integers x, y with $x^3 \neq y^2$.

¹A field embeddable in a finitely generated extension of $\mathbb{Q}_p$.

Remark 14 (abc and the Arithmetic Wall). Szpiro’s conjecture (item 1) is equivalent to abc for elliptic curves and bounds the discriminant by the conductor. This is a bound on how “arithmetically complex” an elliptic curve can be. Since the conductor encodes the ramification of $G_{\mathbb{Q}}$ acting on $T_{\ell}(E)$, Szpiro is a statement about the relationship between the Galois representation and the geometry of E — the arithmetic analogue of the comparison between the Euler characteristic and the complexity of a topological space.

6 Mochizuki’s IUT and the abc Conjecture

Definition 15 (Theta-link and the IUT setup [7]). Mochizuki’s inter-universal Teichmüller theory (IUT) works with *Hodge theaters*: diagrams of anabelian data associated to an elliptic curve E over a number field k . The key construction is the Θ -link: a non-ring-theoretic comparison between two different “copies of the universe” (two Hodge theaters), connected only by their étale fundamental groups. The link is:

$$\Theta^{\pm\text{ell}}\text{NF-Hodge theater} \xrightarrow{\Theta\text{-link}} \Theta^{\pm\text{ell}}\text{NF-Hodge theater}.$$

The two theaters share only their profinite group structure; the ring structures are “forgotten” and reconstructed via anabelian reconstruction algorithms (Mochizuki’s earlier theorems, including Theorem 10).

Theorem 16 (Mochizuki 2012 [7], status disputed). *The IUT formalism yields an inequality*

$$-\log |\Delta_E| \leq -\frac{1}{6} \log |q_E| + (\text{bounded error}),$$

where q_E is the q -parameter of E at its bad primes. This implies the Szpiro conjecture, and hence abc.

Remark 17 (The Scholze–Stix objection). In 2018, Scholze and Stix [8] identified what they consider a gap in IUT: the identification made in Mochizuki’s Corollary 3.12 (the key numerical estimate) appears to conflate distinct objects in the two Hodge theaters. Mochizuki maintains the proof is correct, and the mathematical community remains divided. The proof is accepted for publication in *PRIMS* (2021) but is not universally accepted by the international community.

Remark 18 (IUT as anabelian geometry). IUT’s core move is precisely the anabelian idea: the ring structure of a number field is forgotten, only the Galois group (fundamental group) is retained, and then the ring structure is *reconstructed* via anabelian theorems. The Θ -link is a non-ring-theoretic map that can only be stated in the anabelian world where rings and fundamental groups are cleanly separated. This is a radically different approach to the Arithmetic Wall than Iwasawa theory or perfectoid geometry: instead of constructing Frobenius, IUT abandons the ring structure altogether and works purely with profinite groups. The question is whether the reconstruction is faithful enough to carry the required arithmetic information.

7 The Two Approaches to Rational Points

Theorem 19 (Comparison of approaches to rational points). *The rational points problem ($X(k) = ?$ for a curve of genus ≥ 2 over a number field k) admits two fundamental approaches:*

<i>Approach</i>	<i>Tool</i>	<i>Information used</i>
<i>Cohomological</i>	<i>L-functions, Selmer groups</i>	<i>Abelianization of $\pi_1^{\text{ét}}$</i>
<i>Anabelian</i>	<i>$\pi_1^{\text{ét}}$ directly</i>	<i>Full profinite fundamental group</i>
<i>BSD [10]</i>	<i>Rank of $E(\mathbb{Q})$</i>	<i>$H^1(G_k, T_\ell E)$</i>
<i>Faltings</i>	<i>Finiteness</i>	<i>Galois rep., heights</i>
<i>Section conj.</i>	<i>Exact set $X(k)$</i>	<i>Sections of $\pi_1^{\text{ét}}(X) \rightarrow G_k$</i>

The cohomological approach gives quantitative information (rank, L-value) via abelian data. The anabelian approach gives qualitative information (the exact set of rational points) via non-abelian data. Neither implies the other in full generality.

Remark 20 (The Arithmetic Wall from the anabelian side). The section conjecture would cross the Arithmetic Wall without Frobenius. Over a finite field $\mathbb{F}_q$, rational points correspond to fixed points of Frobenius on the geometric points: $X(\mathbb{F}_q) = X(\overline{\mathbb{F}}_q)^{\text{Fr}_q}$. The section conjecture replaces Fr_q with sections of $\pi_1^{\text{ét}}(X) \rightarrow G_k$. The section is the arithmetic avatar of Frobenius: it says how G_k sits inside $\pi_1^{\text{ét}}(X)$, which is the non-abelian analogue of how Fr_q acts on $\pi_1^{\text{ét}}(X_{\overline{\mathbb{F}}_q})$. The anabelian program and the perfectoid program [11] are thus two strategies for the same wall: perfectoid supplies a local Frobenius at p ; anabelian geometry replaces Frobenius with a non-abelian group-theoretic datum.

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